

THE COMPLETE GUIDE TO RAG EVOLUTION

From Traditional RAG to Graph RAG to Agentic RAG
— Smarter Retrieval. Deeper Reasoning. Real-World Impact.



SWIPE TO LEARN

Explore all 4 essential slides



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Strive For Better Future



AI Coach John
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RAG EVOLUTION CHEAT SHEET

From **Traditional RAG** → **Graph RAG** → **Agentic RAG**

WHAT IS RAG?

Retrieval-Augmented Generation (RAG) combines:



External Knowledge

+



LLM Reasoning

Instead of relying only on model training data, the LLM retrieves relevant information from your data sources before generating a response.



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COMMON RAG USE CASES



Enterprise Knowledge Base



Customer Support Assistant



Internal Company Search



Legal Document Analysis



Healthcare Knowledge Retrieval



Financial Research Assistant



Product Documentation Chatbot

WHY RAG?

Without RAG:

- ❌ Hallucinations
- ❌ Outdated Information
- ❌ No Access To Private Data
- ❌ Expensive Fine-Tuning

With RAG:

- ✅ Grounded Responses
- ✅ Real-Time Knowledge
- ✅ Domain-Specific Answers
- ✅ Lower Cost

THE EVOLUTION OF RAG

Type	Best For	Complexity
Traditional RAG	Document Search	Low
Graph RAG	Relationship Discovery	Medium
Agentic RAG	Multi-Step Reasoning	High

TRADITIONAL RAG

Think:



"Find relevant chunks and answer"

Example:



User: "What is our leave policy?"



Retrieve HR document



Generate answer

Best For:

- FAQs
- Internal Search
- Documentation Assistants



GRAPH RAG

Think:



"Understand relationships"

Example:



User: "Which customers are connected to Project X through Partner Y?"



Traverse graph relationships



Gather connected entities



Generate answer

Best For:

- Knowledge Graphs
- Research
- Fraud Detection
- Complex Relationships



AGENTIC RAG

Think:



"Plan → Retrieve → Reason → Act"

Example:



User: "Compare AWS costs from last quarter and suggest optimization opportunities."



Agent plans tasks



Queries databases



Retrieves reports



Analyzes results



Generates recommendations

Best For:

- AI Copilots
- Research Agents
- Autonomous Workflows
- Multi-Step Tasks



DECISION GUIDE

USE TRADITIONAL RAG IF:



- ✓ Search is enough
- ✓ Documents are independent
- ✓ Fast implementation required

USE GRAPH RAG IF:



- ✓ Relationships matter
- ✓ Multi-hop reasoning needed
- ✓ Connected knowledge exists

USE AGENTIC RAG IF:



- ✓ Complex workflows exist
- ✓ Multiple tools required
- ✓ Planning and reasoning needed



QUICK RULE OF THUMB



Search
→ Traditional RAG



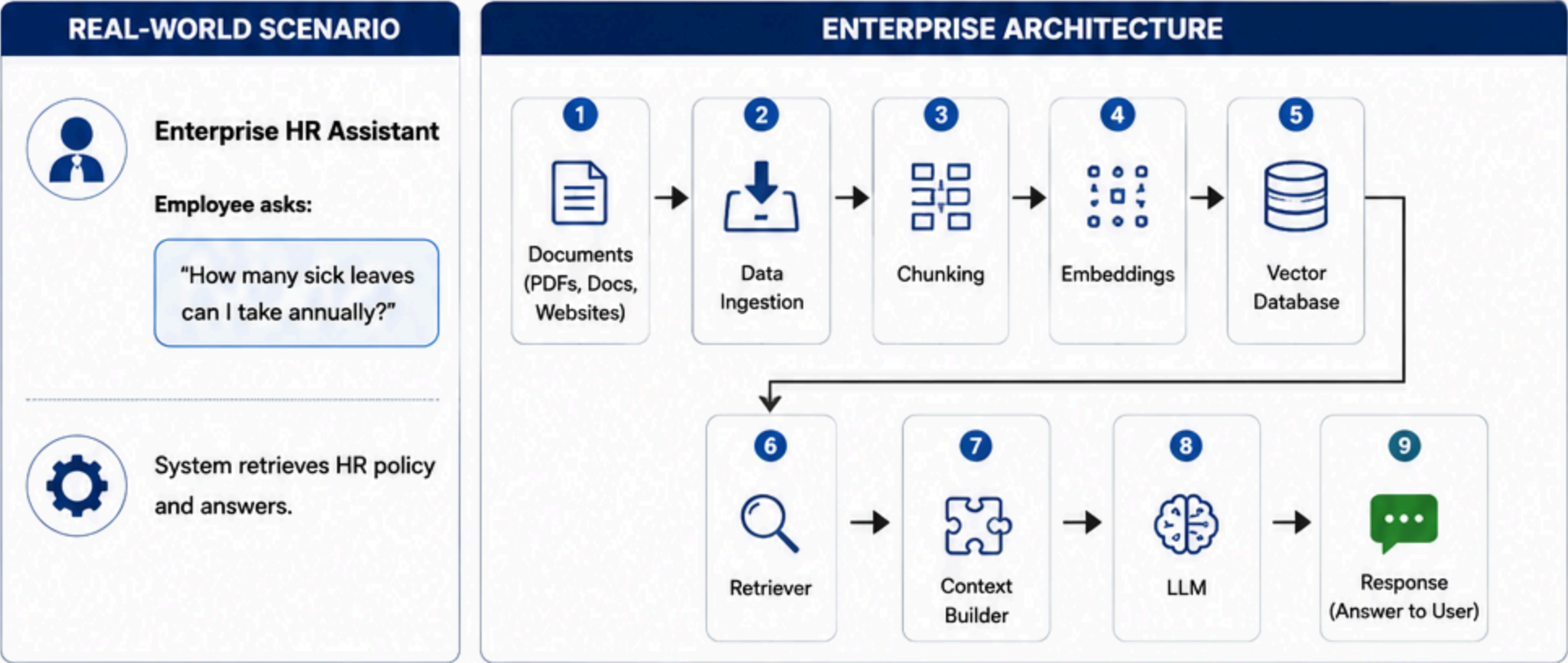
Relationships
→ Graph RAG



Reason + Act
→ Agentic RAG

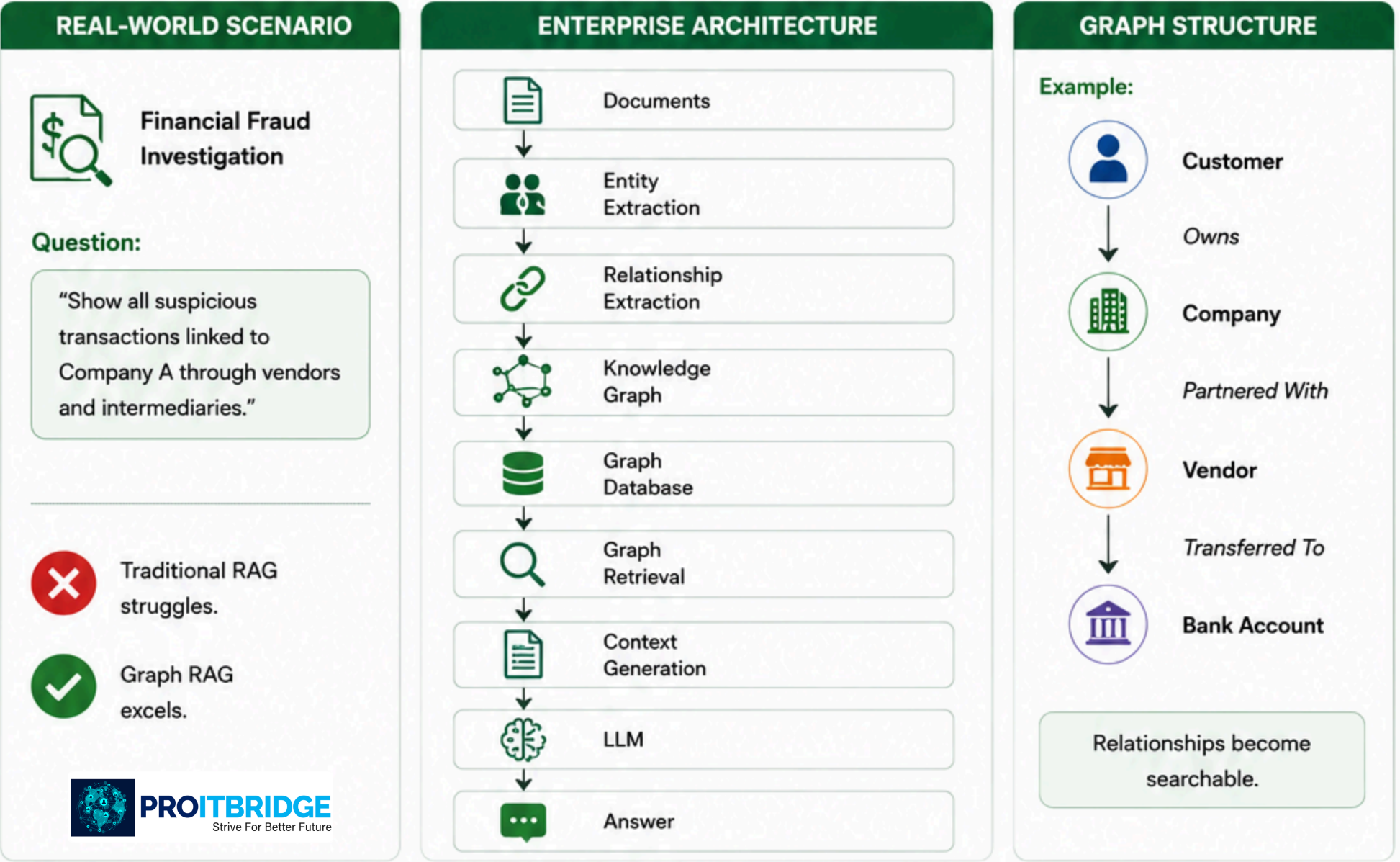
TRADITIONAL RAG DEEP DIVE

How Traditional RAG Works in Enterprise Environments

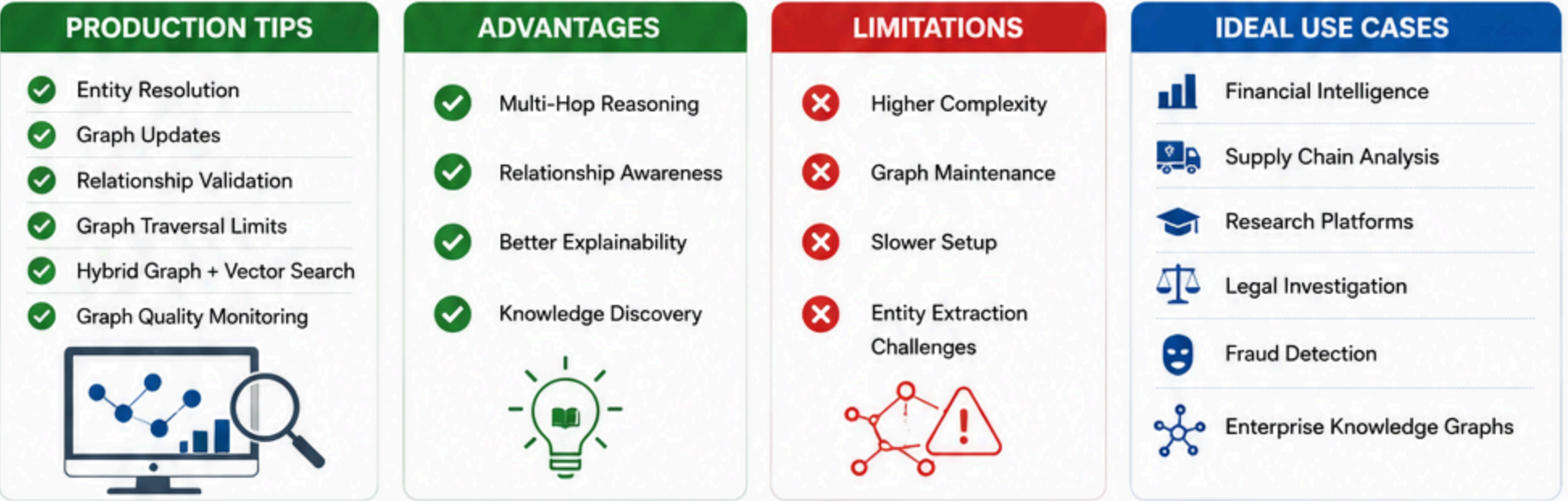
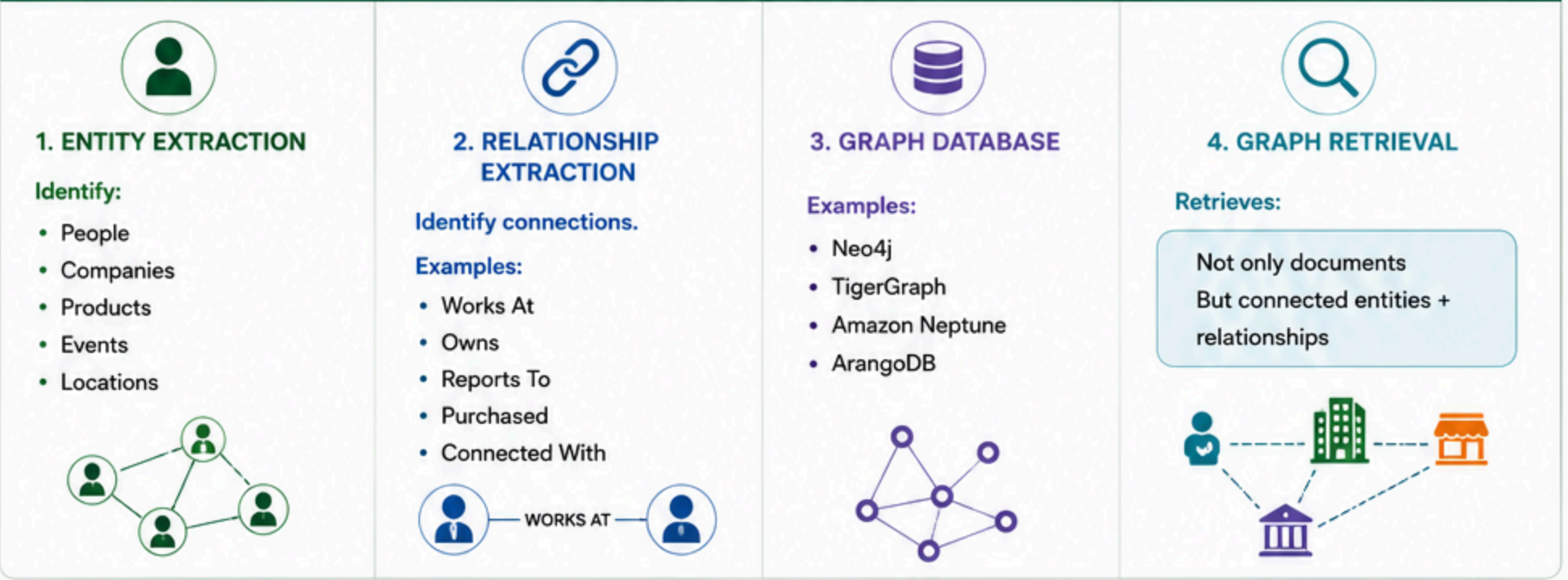


GRAPH RAG DEEP DIVE

Leveraging Relationships for Smarter Retrieval and Reasoning



CORE COMPONENTS



KEY TAKEAWAY

Graph RAG leverages entity relationships and multi-hop connections to deliver deeper insights that traditional retrieval cannot. It is ideal when relationships and context matter.

AGENTIC RAG DEEP DIVE

Autonomous Agents that Plan, Retrieve, Reason and Act

REAL-WORLD SCENARIO

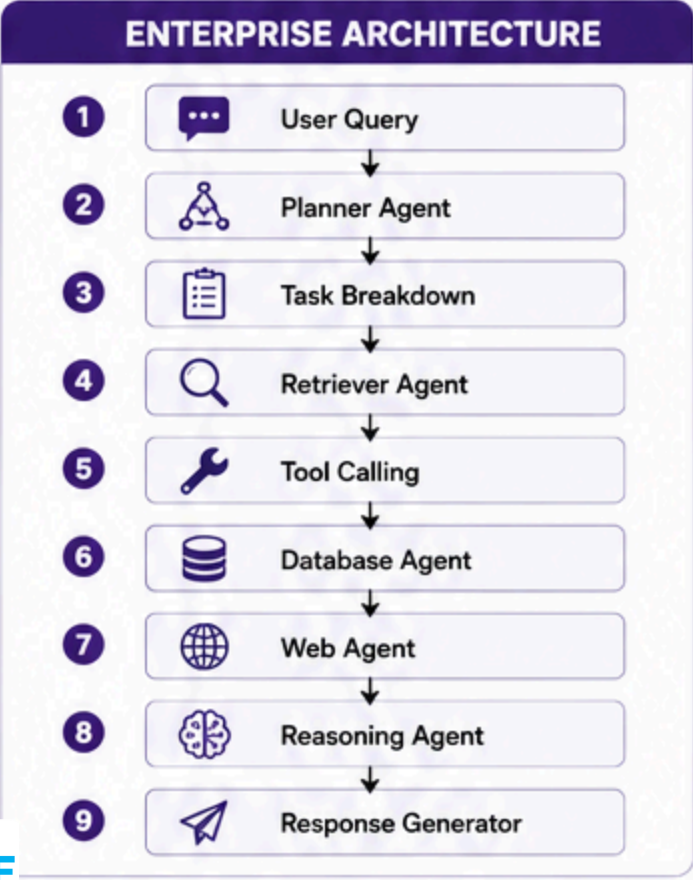
**AI Business Analyst**

Question:

"Analyze Q1 revenue decline, compare with competitors, and recommend actions."

Agent does not simply retrieve.

It thinks and executes.



AGENT WORKFLOW

User Query:

"Find why sales dropped."

Agent:

1



Understand goal

2



Create plan

3



4



5



6



7



8



PRODUCTION BEST PRACTICES

✓ Human Approval Loops

✓ Tool Permission Controls

✓ Cost Monitoring

✓ Agent Observability

✓ Memory Management

✓ Fallback Mechanisms

✓ Guardrails



ADVANTAGES

✓ Complex Problem Solving

✓ Multi-Step Reasoning

✓ Autonomous Execution

✓ Dynamic Workflows



LIMITATIONS

✗ Expensive

✗ Longer Latency

✗ Hard Debugging

✗ Agent Drift Risks



IDEAL USE CASES

AI Copilots

Autonomous Analysts

Research Agents

Operations Automation

Executive Assistants

Enterprise Agent Platforms

FINAL COMPARISON			
Feature	Traditional RAG	Graph RAG	Agentic RAG
 Retrieval	Vector Search	Graph Traversal	Dynamic Retrieval
 Reasoning	Low	Medium	High
 Complexity	Low	Medium	High
 Cost	Low	Medium	High
 Multi-Hop Queries	Poor	Excellent	Excellent
 Tool Usage	No	Limited	Extensive
 Explainability	Medium	High	Medium
 Enterprise Readiness	High	High	Emerging
 Best For	Search	Relationships	Autonomous Tasks

